

A Dual-Tank Model for Real-Time Tracking of Phosphocreatine and Glycolytic Energy Reserves in Cycling

White paper — AnaerobicFuelTanks project Version 0.6 · 2026-07-12 (adds the training-load-partitioning use case — §6.9 — where the decomposition provides information W'_{bal} and the recovery-only null model both lack)

Scope and status (read this once). This is a **physiologically-motivated decomposition of one measured quantity — single-tank W'_{bal} — whose split fraction f_p is assumed, not measured** (power cannot identify the depletion split; §4.2). The two-bar display is precisely characterised, and the characterisation is narrow: **during steady supra-CP effort both bars are affine in W'_{bal}** ($R_p/C_p \approx R_g/C_g \approx W'_{bal} + \text{const}$; §4.2, §4.4), and when the PCr tank is full — most of the time — the PCr bar is likewise an affine transform of W'_{bal} . **All non-redundant content lives in exactly two transients:** (i) the glycolytic *activation ramp* at effort onset, governed by τ_{on} (measured) and τ_{off} (**not** measured), and (ii) the *PCr recovery transient* after hard efforts, governed by τ_p and its new oxidative-headroom gate (§4.2). **There is still no experiment showing the two-bar decomposition is more correct than single-tank W'_{bal}** — that requires the recovery-law head-to-head (§6.6), and at the compartment level it is unfalsifiable in-modality *even in principle* (§6.5, §8). But v0.5–0.6 add **usefulness** arguments the "affine with W'_{bal} " critique does not reach. (i) On real interval rides the PCr bar is informative (below 95%) ~45% of the time and the caps move it up to ~25–60 pts vs a recovery-only null at decision points (§6.8). (ii) For **training** (as opposed to pacing), the *cumulative per-system load* distinguishes an alactic session (~76% PCr) from a glycolytic one (~41%) — information W'_{bal} and the recovery-only null both structurally lack, since the null reports a fixed f_p for every session (§6.9). This second point is the strongest standalone reason to build the second tank. f_p , τ_{off} , and the emergent τ_{dep} remain assumed; τ_g /reconstitution were verified against primary sources (§4.1a). The rest of the paper builds on these; it does not re-litigate them.

Abstract

Endurance-cycling analytics have converged on the Critical Power (CP) / W' framework, in which all work performed above a sustainable power asymptote is drawn from a single finite reserve, W' . This lumps two physiologically distinct anaerobic systems — the fast phosphocreatine (PCr / alactic) system and the slower glycolytic (lactic) system — into one tank with a single recovery time constant. That simplification is convenient but discards the information a rider most wants during hard, variable-intensity efforts: *which*

reserve is being spent, and *how fast* each will come back. This paper proposes a **dual-tank model** that splits W' into a fast PCr reserve and a slow glycolytic reserve, drives both from the instantaneous power trace, and applies system-specific depletion and restoration laws drawn from the bioenergetic and 31P-MRS literature. We are explicit up front about what this is: a **physiologically-motivated decomposition of one measured quantity (W' bal) whose split is assumed, not measured**, and whose extra resolution over single-tank W' bal is real but concentrated in the transients after hard efforts. It generalizes W' bal in depletion and diverges from it — by design — in recovery. The model is deliberately reduced to be computable at 1 Hz on a wrist- or bar-mounted device, and we specify a Garmin Connect IQ implementation that displays live reserve, consumption, and depletion for both systems. We close with a validation strategy — including the one out-of-sample test that would actually earn the second tank — and an explicit statement of the model's assumptions and limits.

1. Introduction

A power meter reports mechanical output, not metabolic cost. The bridge between the two is a model of the body's energy-supply systems. The dominant bridge in cycling — the CP/ W' model — treats supra-CP work as depletion of a single reservoir and sub-CP work as its exponential refill (Skiba et al. 2012). It is power-native, validated, and computationally trivial, which is why it underlies W' bal displays in GoldenCheetah, intervals.icu, and several head units.

Its central simplification is also its central weakness for real-time decision-making: **W' is one tank**. Physiologically, anaerobic capacity is at least two tanks with very different dynamics (Margaria; di Prampero & Ferretti 1999):

- The **phosphocreatine (PCr / alactic)** system — high power, low capacity, **fast** oxidative recovery (seconds to a minute), but with recovery that is pH-sensitive and slows across repeated bouts.
- The **glycolytic (lactic)** system — larger capacity, lower peak power, and **slow** recovery (minutes) that effectively only proceeds at low intensity.

A rider deciding whether to cover the next surge cares which tank is empty. If the PCr tank is full but glycolytic is depleted, a 5-second jump is fine but a 40-second dig is not — a distinction the single-tank W' bal cannot express. The goal of this paper is a model that keeps the power-native, on-device virtues of W' bal while restoring the two-system resolution.

There are two distinct use cases, and they are informed by different outputs. The first — *pacing/racing* — uses the live bars in the moment ("can I cover this attack?"); its value is concentrated in the post-effort transients (§7), and is real but narrow. The second — **training prescription and load monitoring** — uses the *cumulative per-system load* over a session, and it is where the decomposition is most clearly worth more than W' bal (§6.9). The two anaerobic systems carry very different **physiological and recovery costs**: PCr-based (alactic) surges — short, high-power, with adequate recovery — resynthesise in seconds-to-a-minute and can be repeated at high volume with little systemic fatigue or metabolite accumulation; glycolytic surges

accumulate H^+/Pi and clear over many minutes (lactate $t_{1/2} \approx 1366$ s; Ferguson 2010), and cost far more recovery. An athlete who wants to *target* the PCr system — alactic power, repeat-sprint ability — needs to keep efforts short and recoveries full enough that glycolysis stays disengaged, and to know, per session, how much of the anaerobic work was alactic vs glycolytic. W'_{bal} collapses that to one number; the two-tank model does not (§6.9).

Two honest qualifications, stated here rather than buried in §7. First, the split is a **modeling assumption**, not a measurement — power alone cannot identify how W' divides between the two systems (§4.2). Second, the extra resolution is **concentrated in transients**: whenever the fast PCr tank is full — which, given its ~ 27 s recovery, is most of the time — the glycolytic reserve is just an affine rescaling of the existing single W'_{bal} , so the second bar adds decision-relevant information mainly in the ~ 30 – 60 s after a hard effort (which is, however, exactly when covering-the-next-attack decisions are made). The model earns its keep as a *heuristic decomposition*, not as two independently measured reserves.

2. Physiological background

Phosphocreatine (alactic). ATP is buffered by the creatine-kinase reaction ($PCr + ADP \rightarrow ATP + Cr$). Intramuscular PCr (~ 20 – 25 mmol $\cdot$ kg $^{-1}$ wet muscle) is the highest-power energy source and is drawn on instantly at any power transient. Its resynthesis is oxidative and fast: classic alactic O_2 -debt half-time ≈ 30 s (Margaria); ^{31}P -MRS gives recovery time constants $\tau_{PCr} \approx 20$ – 40 s, strongly **pH-dependent** (acidosis slows it) and **biphasic** under high glycolytic load (Harris et al. 1976; Yoshida et al. 2013, PMID 23662804). Recovery also **slows progressively across repeated hard bouts**.

Glycolytic (lactic). Anaerobic glycolysis converts glycogen to lactate, supplying large ATP flux for ~ 10 s– 2 min. Its "fuel level" is best tracked as accumulated muscle/blood lactate — a proxy for H^+ and metabolite accumulation. Contribution decays across repeated sprints ($\approx 40\%$ of ATP in the first of 10×6 s efforts, $< 10\%$ by the last; Sci Rep 2024, DOI 10.1038/s41598-024-78916-z). Restoration is slow (lactic O_2 -debt half-time ≈ 15 min) and, mechanistically, proceeds fastest at low intensity, effectively ceasing above the first lactate threshold (LT1).

The onset that matters. Both systems contribute from the start of a hard effort, but not equally at first: PCr is drawn on instantly (it is the highest-power, immediate buffer), while glycolytic flux takes several seconds to ramp up as glycogen phosphorylase is activated (Parolin et al. 1999) — so the *early* demand is PCr-weighted, shifting toward a shared draw as glycolysis engages. This graded onset — and the very different refill rates — is what the dual-tank model must reproduce (it does so with a parallel, PCr-weighted draw and a glycolytic activation ramp; §4.2), not a strict "PCr first, then glycolytic" hand-off.

What the two "tanks" represent — and what they don't. The "tank" is a deliberate simplification for intuition and on-device display, not a claim that either system is a literal reservoir that drains. The two sides are not even the same *kind* of quantity:

- The **PCr tank** does correspond to a real, depletable substrate store — intramuscular phosphocreatine — so "PCr empty" is close to literal.
- The **glycolytic "tank"** is better understood as **tolerance to accumulating fatigue-related metabolites** than as a fuel gauge. On the ~1–4 min timescale it represents, muscle glycogen is nowhere near exhausted; carbohydrate depletion is a separate, hours-long limiter this model does not address. What forces power down at the limit of a hard effort is the buildup of inorganic phosphate (Pi), H⁺ (acidosis), ADP, and extracellular K⁺ toward the limit of tolerance — consistent with exhaustion at CP/W' coinciding with a reproducible low-PCr / high-metabolite state. (Lactate itself is largely a fuel and a marker, not the cause of force loss.)

This is the physiological reason the two recovery laws differ: PCr resynthesizes quickly, whereas the glycolytic side recovers slowly and only at low intensity because it reflects *clearance and buffering* of accumulated byproducts, which need aerobic metabolism and time — exactly why the model tracks the glycolytic state as an accumulation variable (a muscle-lactate proxy; §4.1) and gates its recovery below LT1. Muscle fatigue is multifactorial and still debated; W' deliberately lumps it into one number for usability. In short: **the PCr tank is a fuel that depletes; the glycolytic "tank" is a byproduct bucket that fills.**

3. Prior models (and why a reduced dual-tank sits between them)

Three model families exist (full survey in `literature-review-anaerobic-models.md`):

- **Family A — CP / W'-balance** (Skiba 2012/2015; Bartram 2018; Froncioni–Clarke). Power-native and on-device-friendly, but **one lumped tank, single recovery τ** . Recovery $\tau_{W'} = 546 \cdot e^{(-0.01 \cdot D_{CP})} + 316 \text{ s}$ cannot represent fast PCr + slow glycolytic simultaneously.
- **Family B — hydraulic tank models** (Morton 1986; Weigend `three_comp_hyd` 2021). **Two explicit anaerobic vessels** (AnF = phosphagen, AnS = glycolytic) plus an aerobic vessel; out-predicts W'bal for intermittent recovery (arXiv 2108.04510). But its 8 parameters are abstract and fit per-athlete by evolutionary computation — heavy for an embedded target.
- **Family C — bioenergetic supply/demand ODEs** (EJAP 2023, PMID 37369795). Explicit alactic / lactic / aerobic metabolic-rate terms, with an efficiency-limited PCr recovery and a power-gated lactate-removal law. Mechanistically ideal but has 17 parameters and needs grey-box fitting.

The gap: Family A is implementable but under-resolved; Families B and C resolve the two systems but are too heavy to run and calibrate on a head unit. The dual-tank model below keeps the two-tank resolution of B/C with the on-device simplicity of A.

To be precise about what "lighter" means here, because it is easy to oversell: the model's core free-parameter count (`CP`, `W'`, `f_p`, `P_p_max`, `g_rate`, `$\tau_p$` , `$\tau_g$` , `$\tau_{on}$` , `LT1`, with `$\tau_{off} = \tau_{on}$` by default — **nine free**, plus two optional realism terms) is essentially the hydraulic model's eight — **the same size, not a smaller model**. The difference is *where the parameters come from*. Only `CP` and `W'` are fitted per athlete (from a standard CP test); the rest are **hard-coded to literature defaults** rather than fit by evolutionary

computation. That buys on-device simplicity and a trivial calibration, at a real cost: the load-bearing defaults (f_p , P_{p_max} , $LT1$, and the recovery τ 's) are exactly the quantities a CP test does **not** determine, so the model's two-system resolution rests on assumptions, not measurements. That is a legitimate identifiability-for-convenience trade — but it is a trade, and the sections below are explicit about which numbers are earned and which are assumed.

4. The dual-tank model

4.1 State and parameters

Two energy reserves (Joules), each with a fixed capacity:

- R_p — **PCr reserve**, capacity C_p
- R_g — **glycolytic reserve**, capacity C_g
- Total anaerobic capacity $W' = C_p + C_g$

The model has **nine free parameters** (plus $\tau_{off} = \tau_{on}$ and two optional realism terms). Only CP and W' come from a CP test; the rest are literature-fixed defaults (see §3 — this is a real count, not a small one):

Symbol	Meaning	Typical default
CP	critical power (W)	from 3-/12-min CP test ($\approx FTP + \text{a few \%}$)
W'	total work above CP (J)	from same test ($\sim 15\text{--}25$ kJ)
f_p	fraction of W' assigned to the PCr tank	0.25 — <i>assumed</i> (physiology, not data; §7)
P_{p_max}	PCr peak power above CP (W), at a full tank	$\approx P_{1s} - CP$ (best 1 s power – CP)
g_rate	glycolytic peak flux as a fraction of PCr peak flux (a ratio)	$0.5 \rightarrow g_{pmax} = g_rate \cdot P_{p_max}$ (flux-ratio note)
τ_p	PCr recovery time constant (s)	27 — within the 31P-MRS τ_{PCr} range (20–40 s), gated by oxidative headroom (§4.2)
τ_g	glycolytic recovery time constant (s)	470 — reconstitution fit (Ferguson 2010 at 20 W $\Rightarrow$ gate ≈ 0.90 ; joint optimum with f_p ; §4.1a)
τ_{on}	glycolytic activation time constant (s)	6 (Parolin 1999)
τ_{off}	glycolytic de -activation time constant (s)	$= \tau_{on}$ — <i>assumed</i> (Parolin measured activation, not de-activation; §7)
$LT1$	first lactate threshold power (W); glycolytic tank recovers below it	measured (a threshold test), <i>not</i> a fixed %CP

Derived: $C_p = f_p \cdot W'$, $C_g = (1 - f_p) \cdot W'$. Initial state $R_p = C_p$, $R_g = C_g$ (full).

Free-parameter count. Nine free (CP , W' , f_p , P_{p_max} , g_rate , τ_p , τ_g , τ_{on} , $LT1$); $\tau_{off} = \tau_{on}$ by default. Only CP and W' are fitted per athlete; the rest are literature-set. That is essentially the hydraulic model's eight — **the same size, but the parameters are literature-set rather than fitted per athlete** (the real trade; §3). A PCr depletion kinetic, $\tau_{dep} = C_p/P_{p_max}$, also falls out of the above (§4.4) and is equally assumed — see §7.

η was removed (was 0.80). The former "recovery efficiency" only rescaled the effective PCr recovery constant to τ_p/η — no sub- C_p asymptote, so not a true efficiency, and degenerate with τ_p when fitted. It silently made the effective constant ≈ 27.6 s while the table advertised 22 s. **$\tau_p = 27$ s is now justified on its own terms:** it sits inside the 31P-MRS τ_{PCr} range of 20–40 s (Yoshida 2013; Harris 1976) — *not* as "22 inflated by η " (that would just re-import the double-count). The **eta** settings key remains, defaulting to 1.0 (identity) with a **deprecation note**, so a future maintainer cannot silently reintroduce the rescale. (The pulmonary- VO_2 recovery half-time of 74 s in Ferguson 2010 is *slower* than muscle τ_{PCr} , because it includes circulatory processes — see the component-mismatch discussion in §4.1a.)

The $g_{pmax} = 0.5 \cdot P_{p_max}$ ratio is not arbitrary: peak glycolytic ATP flux is roughly half peak PCr/creatine-kinase flux in human muscle (di Prampero & Ferretti 1999; the biopsy time-courses of Parolin 1999 / Bogdanis 1996), so PCr is the higher-power system by about 2:1.

Several of these deserve flags up front, because the defaults are doing real work:

- **f_p is assumed on physiological grounds, but now corroborated by the recovery data (§4.1a).** We set **0.25** from physiological alactic-fraction estimates (the PCr system supplies ~ 20 – 30% of anaerobic ATP capacity; di Prampero & Ferretti 1999, Bangsbo 1990), and — once the reconstitution reference protocol's recovery power is known (20 W, near-passive) — the reconstitution curve independently implies $f_p \approx 0.20$ – 0.25 , converging with the physiology. Treat it as a per-athlete calibration target; the supported band is ~ 0.20 – 0.25 (a joint reconstitution fit centres on 0.20).
- **$P_{p_max} \approx P_{1s} - CP$ is the rate at a FULL tank.** PCr flux is not constant: the rate *ceiling* tapers with tank fullness (creatine-kinase equilibrium — flux falls as PCr depletes), so the available PCr rate is $P_{p_max} \cdot (R_p/C_p)$, equal to P_{p_max} only when the tank is full (§4.2). Read $P_{1s} - CP$ as an upper bound: at 1 s glycolysis is already $\sim 15\%$ active, so it mildly over-attributes to PCr. Note this taper is on the *ceiling* only — it does **not** set the submaximal share (§4.2).

4.1a Reconstitution: the source, the calibration, and a contradiction the source raises

The 37% / 65% / 86% W' -recovery at 2 / 6 / 15 min (half-time 234 s) is **Ferguson et al. 2010** (*J Appl Physiol*, DOI 10.1152/jappphysiol.91425.2008), reviewed by Chorley & Lamb 2020, who state it was measured "when recovering at a **nominal 20 W**." Three things follow.

- **τ_g is calibrated to ~ 470 s, not confounded (but not exactly "passive").** The confound of review 3 — the LT1 gate makes the effective constant $\tau_g/gate$, so the fit depends on recovery power — is

resolved by the 20 W figure. But 20 W is not quite $\text{gate} = 1$: with $\text{LT1} \approx 0.8 \cdot \text{CP} \approx 200 \text{ W}$, $\text{gate} = (200-20)/200 = 0.90$, so a model reproducing Ferguson needs $\tau_g \approx 470 \text{ s}$ ($\pm$ a few % on the cohort's unreported LT1), **not 520**. The default is corrected to **470**.

- **f_p is consistent with the curve, not independently corroborated by it.** With recovery power fixed, the curve still constrains a **ridge in (f_p, τ_g)** , not f_p alone: $(f_p \ 0.25, \tau_g \ 520)$ and the true joint optimum **$(f_p \ 0.20, \tau_g \ 470)$** (SSE $\approx 10.7 \text{ pts}^2$) both fit 37/65/86 to within ~ 3 –5 pts. So the honest statement is *one* physiological estimate of f_p (di Prampero & Ferretti 1999, alactic O_2 -debt $t_{1/2} \approx 30 \text{ s}$; Bangsbo 1990, $\sim 20\%$ alactic) **plus a reconstitution curve consistent with it** — the physiology is what selects $f_p \approx 0.25$ on the ridge. We keep $f_p = 0.25$ as the physiological midpoint but note the recovery data alone would prefer ~ 0.20 . (*An earlier draft called this "two independent lines converge"; that overstated the independence — corrected here.*)
- **Ferguson's own components contradict the model's identification — and this is the strongest single datum in the paper.** Ferguson reports the two recovery channels underlying W' reconstitution: $\text{VO}_2 \ t_{1/2} = 74 \text{ s}$ (a proxy for oxidative/PCr resynthesis) and blood-lactate $t_{1/2} = 1366 \text{ s}$. Put those beside the model's component half-times:

Component	Model	Ferguson channel	ratio
fast	$\tau_p = 27 \text{ s} \rightarrow t_{1/2} \ 18.7 \text{ s}$	$\text{VO}_2 \ 74 \text{ s}$	4.0× slower
slow	$\tau_g = 470 \text{ s} \rightarrow t_{1/2} \ 326 \text{ s}$	lactate 1366 s	4.2× slower

Both of the model's components are $\sim 4\times$ too fast, yet the aggregate 234 s half-time reproduces. That is the signature of fitting two exponentials to a *sum* without checking them against the *components* — the identifiability problem this review series has circled, now with a number. And Ferguson's own conclusion is blunter still: W' reconstitution is "**not a unique function of phosphocreatine concentration or arterial [lactate], and it is unlikely to simply reflect a finite energy store that becomes depleted.**" *The primary source used to calibrate the recovery law explicitly denies that the recovery decomposes the way the model decomposes it.* We host that here rather than resolve it by citation-picking. Two defences, and their limits: (i) VO_2 off-kinetics is **not** PCr resynthesis — it includes slower circulatory processes — and muscle 31P-MRS τ_{PCr} of 20–40 s does support $\tau_p = 27$; so τ_p is defensible against the 31P literature while being $4\times$ off against the channel measured in the very protocol that produced the recovery curve. (ii) The channel *separation* (74 s vs 1366 s, $\sim 18\times$) genuinely supports a bi-exponential structure, even if the channel *values* are not the tanks. The tension is the finding; the two-tank recovery is a heuristic that fits the aggregate, not a measured decomposition of it.

4.2 The 1 Hz update

Let P be instantaneous power (W) and $\Delta t = 1 \text{ s}$. Define the demand relative to aerobic supply as $\Delta = P - \text{CP}$.

Case 1 — depletion ($\Delta > 0$, above CP). Demand $\text{need} = \Delta \cdot \Delta t$ (J) is met by **both systems in parallel**. This matters: glycolytic ATP supply is not instantaneous. Glycogen phosphorylase (the rate-limiting

glycolytic enzyme) transforms to its active form over the first several seconds of maximal effort — Parolin *et al.* (1999) measured it rising from 12% at rest to 47% by 6 s — while phosphocreatine is the immediate buffer (a fresh 30 s sprint drains it to ~17%; a 10 s sprint leaves ~20–30%; Bogdanis *et al.* 1996); both contribute from the onset, not in sequence (González-Alonso *et al.* 2000). We model this with a **glycolytic activation** term $g \in [0, 1]$ that ramps first-order with a time constant $\tau_{\text{on}} \approx 6 \text{ s}$ while above CP and relaxes back during recovery:

```
g ← g + (1 - g) · (1 - exp(-Δt/τ_on))      # glycolytic activation ("inertia")

# --- SHARE of submaximal demand: capacity-proportional (NOT rate-proportional) ---
w_p = C_p                                  # at g=0 → PCr covers all; at g=1 → PCr cov
w_g = C_g · g
share_p = need · w_p / (w_p + w_g)          # (guard: if w_p+w_g ≈ 0, share_p = need)
share_g = need - share_p

# --- RATE CEILING: peak flux, PCr tapered with fullness. Governs MAXIMAL efforts only
rate_p = P_p_max · (R_p / C_p)              # (guard: C_p > 0)
rate_g = g_rate · P_p_max · g

take_p = min(share_p, R_p, rate_p·Δt)        # each tank capacity- AND rate-limited
take_g = min(share_g, R_g, rate_g·Δt)
unmet = need - take_p - take_g
# shortfall (a tank empty or rate-capped) spills to the partner, then to a deficit:
take_g += min(unmet, R_g - take_g, rate_g·Δt - take_g);  unmet -= ...
take_p += min(unmet, R_p - take_p, rate_p·Δt - take_p);  unmet -= ...
R_p -= take_p;  R_g -= take_g
D += unmet                                  # DEFICIT (debt): supra-CP work the caps co

if unmet > 0:  rate_limited = true           # producing power beyond the tanks' flux -
```

Why the share and the ceiling are different objects (this is the fix that ended a three-round patch cycle).

The rate ceiling — $P_{\text{p_max}}$, g_{pmax} — encodes "PCr is the higher-power system." That is true at **maximal** effort, where both systems are flux-limited, and it is where PCr dominance belongs. It is the **wrong** object for apportioning **submaximal** supra-CP demand, where neither system is near its flux limit and the sharing is set by metabolic control, not peak-flux ratios. Earlier revisions used one rate-weighted rule for both jobs, which forced the PCr trajectory to be a fixed, convex, intensity-*invariant* function of W' -spent (PCr at ~9% by the midpoint of *any* hard effort). Weighting the **share by capacity** and keeping the taper on the **ceiling** fixes it with no cost in the sprint case (there the caps and spill dominate and the share rule never binds — verified bit-identical):

- **Submaximal:** both tanks drain $\approx$ proportionally to capacity, so $R_p/C_p \approx R_g/C_g \approx W'_{bal}$ and both reach their nadir *together* at exhaustion — matching the reproducible-metabolic-milieu picture in §2. Honestly, this means during a steady supra-CP effort the two bars track W'_{bal} and each other; they diverge only in the activation ramp and in recovery. That is not a defect — it is the model behaving the way the Scope box and §7 already say it does (extra resolution lives in transients).
- **Maximal:** the tapered ceiling makes R_p decay $\sim$ geometrically to its nadir at exhaustion (a ~ 10 s all-out sprint leaves $R_p \approx 20\text{--}30\%$, matching Bogdanis 1996; §4.4), and PCr dominance emerges from the ceiling, not the weight.
- **Energy conserved even when a cap binds.** Residual $unmet$ is banked in a **deficit D** (standard W'_{bal} permits a negative balance), so $(R_p + R_g - D)$ drops by exactly $\Delta \cdot \Delta t$ per second and §6.2 holds. D clears only below LT1 (Case 2) — it is supra-cap byproduct load and must respect the same intensity gate as the glycolytic tank. Note the deficit preserves the *aggregate* but not the *split*: work booked to D never drains a tank, so if D is large the two bars read slightly full. D is an accounting term for the rare supra-flux case, usually signalling a **stale P_{1s}** , not fatigue.

On identifiability (and consistency with §7). The parallel draw is the physiologically faithful choice for the *live display*, and τ_{on} is a literature-set constant (Parolin 1999), not a fitted one. But we are careful not to over-claim: the depletion split is **not** identifiable from power at all — total draw above CP is all the pedals see, and any (f_p, g_{pmax}) that sums to the same total is observationally equivalent. In *principle* the bi-exponential **recovery** of W' constrains the split (a fast PCr component + a slow glycolytic component); in *practice* that fit is ill-conditioned (§6.3 shows the fast component is invisible at standard reconstitution sampling), so f_p ends up **assumed and weakly constrained**, not recovered. This is the §7 position, and it is the one to trust — the reserves are a plausible decomposition, not measured latent state.

Case 2 — restoration ($\Delta \leq 0$, at/below CP). Recovery "headroom" is $CP - P$. Each tank refills exponentially toward full; glycolytic refill is gated by intensity.

```
g ← g · exp(-Δt/τ_off)                # glycolytic DE-activation (τ_off =

# PCr: fast, oxidative – resynthesis needs aerobic ATP ABOVE the ride's own demand, so
# gated by the OXIDATIVE HEADROOM (CP – P): full at rest, near-arrested at CP.
gate_p = max(0, (CP – P) / CP)
R_p += gate_p · (C_p – R_p) · (1 – exp(-Δt/τ_p)) # τ_p from 31P-MRS τ_PCr (§4.1)

# Glycolytic tank AND the deficit clear only below LT1 (a MEASURED power, not a %CP)
if P < LT1:
    gate = (LT1 – P) / LT1                # 0 at LT1, → 1 toward zero power
    R_g += gate · (C_g – R_g) · (1 – exp(-Δt/τ_g))
    D    -= gate · D · (1 – exp(-Δt/τ_g))    # debt clears on the same gate as R_
```

The PCr recovery gate is new in v0.5, and it fixes the model's headline use case. Until now PCr recovered at the same rate whether the rider was stopped or holding 99% of CP — so the "punch" bar went green while the rider was still under load, in exactly the "*can I cover this next attack?*" scenario the field exists for (sitting in a bunch between attacks is recovery at high sub-CP power). PCr resynthesis is oxidative and needs spare aerobic capacity, which vanishes as $P \rightarrow CP$; the $\text{gate}_p = (CP - P)/CP$ term encodes that. At $P = 0$ it is identical to the old law ($\text{gate}_p = 1$), so nothing in depletion or the reconstitution fit moves; at tempo it does the physiologically obvious thing. Verified: after a 20 s / 700 W effort ($PCr \rightarrow 23\%$), 60 s of recovery now restores PCr to **87% at 0 W, 46% at 0.8·CP, and 25% at 0.99·CP** — versus a flat 92% at every recovery power before the gate.

The g de-activation is what makes the activation ramp re-fire on each interval of a repeated-bout set (without it, g would ratchet to 1 on the first surge and every later bout would start at a flat split — the ramp inert for the rest of the ride). **$\tau_{\text{off}} = \tau_{\text{on}}$ is an assumption, and a load-bearing one:** Parolin (1999) measured phosphorylase *activation*, not *de-activation*, which is a separate quantity. If de-activation runs on a minutes timescale rather than seconds, the ramp would not re-fire on 30 s recoveries and the repeated-sprint behaviour changes materially — so τ_{off} is flagged alongside f_p , τ_g (§7).

LT1 is the rider's **first lactate threshold in watts**, from a threshold test — *not* a fixed fraction of CP. LT1 and CP vary independently (LT1 is roughly 65–85% of CP depending on training status), and this gate decides whether the glycolytic tank refills at all during tempo/endurance riding, so getting it wrong there is consequential. Where only CP is known, $LT1 \approx 0.80 \cdot CP$ is a fallback, but it is the weakest default in the model and should be replaced by a real measurement.

Optional realism (recommended, off by default): - *pH-slowed / fatiguing PCr recovery*: scale τ_p upward as the glycolytic tank is emptier, e.g. $\tau_{p_eff} = \tau_p \cdot (1 + k \cdot (1 - R_g/C_g))$, capturing the observed slowing across repeated bouts. - *Aerobic ramp*: replace the hard CP boundary with a first-order aerobic supply $A(t)$ that rises toward $\min(P, CP)$ with $\tau_{aer} \approx 25$ s, and use $need = (P - A) \cdot \Delta t$. This reproduces the onset "oxygen deficit" that transiently loads the anaerobic tanks even below CP.

4.3 Reported quantities

- **PCr reserve** = R_p / C_p (0–100%) — how much "punch" is left.
- **Glycolytic reserve** = R_g / C_g (0–100%) — how much "sustained dig" is left.
- **Consumption rate** (per system, W) — $\text{take}_p/\Delta t$, $\text{take}_g/\Delta t$, the live draw on each tank.
- **Combined W'_{bal}** = $(R_p + R_g - D)/W'$ — the deficit term keeps this **energy-conserving in depletion** (it drops by exactly $\Delta \cdot \Delta t$ per second) so it matches a single-tank W'_{bal} reference on above-CP segments. It intentionally **diverges in recovery** (bi-exponential + LT1 gate; §4.4/§6.2), so it is *depletion-compatible*, not identical everywhere.
- **Depletion / exhaustion flag** — the rider is producing power beyond the tanks' rate/capacity (D growing), or both reserves are at/near zero.

4.4 Why this behaves correctly

- A short, very hard surge draws mostly from R_p (glycolysis has not yet ramped in), and R_p refills after — *at a rate set by recovery intensity* (§4.2 gate). A maximal ~10 s sprint leaves $R_p \approx 20\text{--}30\%$, **not empty**, from the tapered ceiling's emergent depletion constant $\tau_{\text{dep}} = C_p/P_{p_max}$. **But τ_{dep} is not a fixed 7 s** — §7 notes it swings ~4–13 s across riders, so the 10 s residual ranges from ~10% (fast riders) to ~46% ($\tau_{\text{dep}} \approx 13$ s). The central case matches Bogdanis 1996 (PCr $\approx 17\%$ at 30 s); the high tail does *not*, which is an argument for a *measured* PCr depletion constant (the re-architect fork, §7), not an emergent ratio.
- A sustained supra-CP effort draws from **both** tanks, declining together to their nadir at exhaustion. The drawdown is **front-loaded (mildly concave), not linear** — the activation ramp makes the early draw PCr-weighted (450 W: 68/42/18/1 at quarter-TTE intervals). During such steady efforts **both bars are affine in W'_{bal}** (§4.2), so they carry no information single-tank W'_{bal} lacks *there*; their only non-redundant content is the ramp offset (set by $\tau_{\text{on}} / \tau_{\text{off}}$) and the post-effort PCr recovery transient. This is stated as a limitation, not a feature.
- **A falsification test (not a validation), and its real cost.** For a constant supra-CP effort, R_p/C_p is *asymptotically* a function of the fraction ϕ of W' spent — but only for efforts long relative to τ_{on} and below the ceiling regime; the activation ramp adds a genuine second intensity dependence (a hard effort reaches a given ϕ faster, with g lower, so PCr is *lower* at that ϕ). So the test "hold different supra-CP powers to exhaustion and compare PCr-at-% W' -spent" would report *intensity-dependence* for a correct model, and a naive read would wrongly "falsify" it. Framed correctly it is a **falsification** test (pass $\Rightarrow$ the bar is a deterministic restatement of W'_{bal} in depletion, i.e. redundant; fail $\Rightarrow$ the architecture is wrong) — lose-lose for the *depletion* claim, which is the point. And it is **not** "testable in an afternoon": checking PCr in-cycling needs 31P-MRS, which §6.5 says is unavailable in this modality. It requires a surrogate (post-exercise sampling, or knee-extension 31P-MRS with a cross-modality transfer assumption; §8).
- **Synthetic battery (v0.5).** Parameters: $CP = 255$ W, $W' = 20$ kJ, $f_p = 0.25$, $P_{p_max} = 690$ W ($g_{\text{rate}} = 0.5$), $\tau_p = 27$, $\tau_g = 470$, $\tau_{\text{on}} = \tau_{\text{off}} = 6$, $LT1 = 204$ W.

Test	Result
Sustained 450 W — PCr at 25/50/75/100% of TTE	68 / 42 / 18 / ~1% (front-loaded, nadir at exhaustion)
Both tanks at exhaustion	PCr ~1% / GLY ~0% (empty together)
945 W (= P_{1s}) 10 s sprint — PCr residual	23% (ceiling-dominated; share rule doesn't bind)
6×[10 s@700 W / 30 s@150 W] — PCr at each bout end	49 → 17 → 10 → 8 → 8 → 7% (gated recovery: 150 W is high sub-CP, so little refill)
1200 W hold (caps bind) — combined W'_{bal} conservation	leak = 0 J
PCr recovery after 700 W/20 s, 60 s @ {0, 128, 204, 242, 252} W	87 / 68 / 46 / 30 / 25% (headroom gate)

Test	Result
Debt repayment at 0.9·CP (above LT1)	D unchanged (LT1-gated)

- **On "generalizing W'bal" — a careful claim.** In *depletion* the model is a faithful generalization: $(R_p + R_g - D)$ falls by exactly $\Delta \cdot \Delta t$ per second (cap-binding or not), so the summed draw above CP matches standard W'bal. In *recovery* it is deliberately **different** (bi-exponential, LT1-gated), so it does not reduce to Skiba's single $\tau_{W'}$. It is a generalization in **structure and depletion behaviour**, not a strict superset. (We drop the earlier $f_p \rightarrow 0$ argument: with the fullness taper $C_p = f_p \cdot W' = 0$ is a division by zero, guarded in code — the limit is not meaningfully computable and does not add anything.)

5. Real-time on-device implementation (Garmin Connect IQ)

The model's per-second cost is a handful of multiplies and one `exp()` per tank — trivial for a head unit. A companion document, `connectiq-app-spec-and-prompt.md`, gives the full technical specification and a ready-to-use build prompt. Summary of the target design:

- **App type:** a custom full-screen **Data Field** (`Toybox.WatchUi.DataField`), which the head unit calls once per second — a natural fit for the $\Delta t = 1\text{ s}$ update.
- **Input:** `Activity.Info.currentPower` inside `compute(info)`.
- **Configuration:** `CP`, `W'`, `f_p`, `P_p_max` (a per-athlete calibration target — §6.7), **LT1** (a measured threshold, not a %CP default — §4.2), and the recovery constants exposed via Connect IQ app settings (`Application.Properties`), so a rider enters them from Garmin Connect.
- **Display:** two vertical bar gauges (PCr and glycolytic reserve) with numeric % and color bands (green → amber → red). *Per-system live consumption in W is a "modelled share," not a reading* — when the ceiling is slack it is just $(\text{power} - \text{CP})$ split by a fixed fraction (§7), so it is labelled as such or omitted.
- **Recording:** write both reserves and consumption to the FIT file via `FitContributor` fields so they sync to Garmin Connect / intervals.icu / Strava for post-ride analysis.
- **Footprint:** a handful of scalar state variables (`R_p`, `R_g`, the activation `g`, the deficit `D`), no arrays — well within Connect IQ memory budgets. Per-second cost is still a few multiplies and one `exp()` per tank.

6. Validation strategy

The model is a hypothesis; it must be checked against data before its numbers are trusted. The honest difficulty is that the model's *headline feature* — a separate PCr reserve tracked in real cycling — is the

hardest thing to validate, so we separate tests that only check the plumbing from the one test that would actually earn the second tank.

1. **Face validity / unit tests** — synthetic power traces (single sprint, repeated sprints, supra-CP hold, sub-CP recovery) should produce the qualitative behaviours in §4.4.
2. **Backward compatibility — depletion only.** The combined balance $(R_p + R_g - D)$ reproduces a reference W' bal implementation (Froncioni–Clarke) on the same trace **in depletion** — it drops by exactly $\Delta \cdot \Delta t$ per second, and the deficit term D preserves that even when a rate cap binds. It will **not** match in recovery, by construction (§4.4), so this test must be scoped to above-CP segments — a test that expected agreement everywhere is one the model is designed to fail.
3. **Reconstitution targets — what the curve constrains, and what it cannot.** Combined recovery after a full W' depletion is compared to the published curve (37 / 65 / 86% at 2 / 6 / 15 min; Ferguson et al. 2010, at 20 W recovery). (a) **The curve is blind to the fast tank** — a $\tau_p \approx 27$ s process is $1 - e^{(-120/27)} \approx 99\%$ complete by the first (2 min) sample, so it validates *glycolytic* recovery and the size of the PCr offset, **not** PCr kinetics (for those you need 10/20/30/45/60 s samples; test 4). (b) **The curve fixes τ_g and constrains a (f_p, τ_g) ridge, not f_p alone.** With the 20 W recovery power known (gate ≈ 0.90), the fit gives $\tau_g \approx 470$ and the joint optimum $(f_p \approx 0.20, \tau_g \approx 470)$; f_p is *selected on that ridge by physiology*, not by the curve (§4.1a). So the curve is **consistent with $f_p \approx 0.20\text{--}0.25$** , not an independent measurement of it. (*This supersedes a v0.3 statement — since deleted — that computed an implied $f_p \approx 0.13$ at the old $\tau_g = 360$ s; that value was a fossil of the superseded default.*)
4. **Fast-recovery kinetics (the missing piece).** The load-bearing test for τ_p is a repeated-bout protocol with **early** recovery sampling — e.g. all-out efforts separated by 10/20/30/45/60 s — and the between-bout power recovery correlated with modelled PCr recovery (as Bogdanis 1996 did for PCr). Standard reconstitution data cannot supply this; it must be measured deliberately.
5. **System-specific truth (lab) — and its hard limit.** Blood-lactate kinetics can corroborate the glycolytic tank, but the PCr tank's gold standard, **31P-MRS, is essentially unavailable in real cycling**: the measurement requires the muscle to be inside a magnet, so it exists only for small-muscle knee-extension or immediate post-exercise, not whole-body cycling. The model's headline novelty therefore **cannot be checked against the gold standard in the modality it targets** — a limitation to state plainly, not paper over.
6. **The test that earns the second tank — and what it actually tests.** Because the model is a near-superset of single-tank in depletion, "it fits better" is guaranteed and is **not** evidence the compartments are real. The decisive test is a **head-to-head against single-tank W' bal on an out-of-sample intermittent-tolerance prediction** (the protocol class where the hydraulic model beat W' bal, arXiv 2108.04510). But be precise about what it probes: since this model's *depletion* is single-tank by construction, its **only** lever to beat single-tank is the **recovery law** (bi-exponential + LT1 gate). So **test 6 is a test of the recovery law, not of the two-compartment hypothesis** — the framing "two tanks → better prediction" overstates what a win would show. This also sharpens the protocol: choose interval recovery-valley powers that **straddle LT1**, where the dual-tank and single-tank recovery laws diverge most, rather than a generic interval set.

7. **Field calibration — scoped honestly.** Fit **P_pmax** and the **τ** 's per athlete to maximal and repeated-bout tests. **f_p** is *not* a routine calibration target: §4.2 shows the depletion split is unidentifiable from power, and only the test-4 class (all-out efforts with early recovery sampling) can constrain it — and even then the fit is ill-conditioned. Also: any single-tank W'bal comparator in test 6 must have **its own τ_W** fit to the same data, or the dual-tank wins trivially by having more recovery freedom and the win means nothing.

6.8 First empirical results (v0.5) — usefulness, on real rides

Three reviews converged on the point that the paper validated *plausibility* but never the two cheap, data-in-hand questions that decide whether the feature is worth building. Both were run on six real interval/VO₂max sessions (the model's best case — a steady endurance ride would score far lower):

- **How often is the PCr bar informative?** The bar carries information beyond single-tank W'bal only when the PCr tank is not full. Across the six rides, **R_p** is below 95% for ~45% of ride time (range 20–79%) — well above the ~3% ("curiosity") and ~20% ("real") thresholds a reviewer proposed. On interval sessions the second bar is live nearly half the time. (*Caveat: these are hard interval files; on a steady endurance ride PCr is full most of the time and the figure would be small.*)
- **Does the depletion machinery change the display, vs the recovery-only null model?** Running the full model and the null (§7) on the same traces, the PCr bar diverges by a mean of ~6 pts but up to ~25–60 pts, with the maxima at the **starts of recovery valleys** — right after attacks, the decision-relevant window. That is well above the ~5-pt "a rider could act on it" threshold, so **on interval sessions the caps earn their keep**: they set the post-effort PCr level the recovery law then acts on, and a null model would read materially differently exactly when it matters. This is the first evidence that the depletion-side machinery is not merely inert plumbing — though only in the maximal regime (§7 discusses where it is, and is not, doing work).

Neither result validates the *compartments* (that is §6.6, and at the compartment level unfalsifiable in-modality — §6.5, §8). They answer a narrower, real question — *is the two-bar display worth showing, and does the model's complexity change what it shows* — with data instead of assertion, and the answers are yes and yes, for interval training.

6.9 Training-load partitioning — a use case the reviews evaluated past

Every review assessed the model as a **pacing** aid and correctly found the live bars affine with W'bal outside transients. But there is a second use case they did not weigh, and it is where the decomposition is most clearly worth more than W'bal: **training prescription and load monitoring**. The relevant output here is not the live bar but the **cumulative per-system load over a session** — how many kJ came from the alactic (PCr) vs the glycolytic system — which the field already records (**PCr_depleted_kJ** , **GLY_depleted_kJ**). This matters because the two systems carry asymmetric recovery costs (§1): alactic work is cheap and repeatable; glycolytic work is expensive and self-limiting. A coach building a PCr-targeted block (alactic power, repeat-

sprint ability) wants sessions that stay alactic; one building lactate tolerance wants the opposite. **W'bal cannot tell these apart — it is one number.**

The model can, because the **activation ramp** makes the per-system split depend on *effort structure*, not just f_p : short efforts keep g low (PCr-weighted); sustained efforts let $g \rightarrow 1$ (glycolytic-weighted).

Simulated per-system load for three archetypal sessions (default parameters):

Session	PCr load	Glyc load	PCr % of anaerobic load	W'bal	recovery-only null
10 × [6 s @ 900 W / full recovery] — alactic	26.8 kJ	8.7 kJ	76%	one number	25%
5 × [60 s @ 360 W / short recovery] — glycolytic	12.1 kJ	17.7 kJ	41%	one number	25%
3 × [8 min over-under / sustained] — glycolytic	12.7 kJ	18.3 kJ	41%	one number	25%

Two things stand out. First, the model **discriminates** the alactic session (76% PCr) from the glycolytic ones (41%) — the qualitative signal a training plan needs. Second, this is a use case where **the depletion-side machinery earns its keep and the recovery-only null model does not**: the null depletes both tanks proportionally to capacity regardless of effort structure, so it reports a fixed f_p (25%) for every session and is blind to exactly the distinction that matters here. The discriminating content is the activation ramp — precisely the τ_{on} / τ_{off} term the reviews identified as the depletion phase's only non-redundant information (§7), now shown to carry a concrete, actionable signal.

Honest limits: the *absolute* per-system kJ scale with the assumed f_p , so the exact percentages are soft; what is robust is the **ordering and separation** between session types, which the ramp drives. And this is still a training-*design* signal, not a validated training *outcome* — it says a session was alactic- or glycolytic-dominant, not that partitioning training by it improves adaptation (an intervention study, not a modelling result). But unlike the pacing case, here the two-bar model provides information W'bal and the null model both *structurally lack* — which is the strongest standalone argument in the paper for building the second tank at all.

7. Assumptions and limitations

- **The PCr/glycolytic split f_p is assumed (physiology), and *consistent with* — not corroborated by — the recovery data.** Power cannot identify the depletion split (§4.2). We default to **0.25** from physiological alactic-fraction estimates (~0.20–0.30; di Prampero & Ferretti 1999, Bangsbo 1990 ~20%). The reconstitution curve constrains a (f_p, τ_g) ridge, not f_p , and physiology is what selects f_p on it (§4.1a) — so the two "lines" are not independent, and the recovery data alone would prefer ~0.20. Supported band ~0.20–0.25; a per-athlete target; outputs are sensitive to it.

- **The reserves are a decomposition, not latent state.** The two bars are a physiologically-motivated split of one measured quantity (W'_{bal}), not two independently measured reserves. When the PCr tank is full — which, given $\tau_p \approx 27\text{ s}$, is most of the time except the $\sim 30\text{--}60\text{ s}$ after a hard effort — the glycolytic bar is an affine transform of the existing single W'_{bal} ($W'_{bal} = f_p + (1-f_p) \cdot R_g/C_g$), so it carries no new information there. The PCr bar adds decision-relevant content **only** in those post-surge transients. Those transients *are* where race decisions concentrate (can I cover this attack right after the last one?), which is the case for the display — but the honest framing is a heuristic decomposition whose split is assumed, not "the two numbers a rider races on" as if both were measured.
- **The PCr depletion rate is also assumed — and it was invisible until v0.4.** The tapered ceiling gives an emergent depletion constant $\tau_{dep} = C_p/P_{p_max}$, tied to *no* measured PCr depletion kinetic; it is the ratio of three parameters and swings $\sim 3\times$ across plausible riders ($\sim 4\text{--}13\text{ s}$). §7 was scrupulous about the assumed *recovery* constants and silent about this equally load-bearing *depletion* one. It should be sanity-checked against literature PCr depletion half-times, not left to fall out of f_p , W' , and a 1 s sprint power.
- **τ_g is calibrated to ~ 470 (confound resolved); τ_{off} remains assumed.** Ferguson recovered at $\sim 20\text{ W}$ (gate ≈ 0.90), so $\tau_g \approx 470$ — not 520, and not the 260 a soft-pedal recovery would have implied (§4.1a). $\tau_{off} = \tau_{on}$ is still an assumption (Parolin measured activation, not de-activation; §4.2) and, per Reviewer 2, it is the *only* parameter carrying the depletion-phase novelty: since both bars are affine in W'_{bal} in steady effort, the non-redundant depletion content is the activation-ramp offset, which is a function of τ_{on} (measured) and τ_{off} (not).
- **LT1 should be measured, not derived from CP.** The $0.80 \cdot CP$ fallback will be wrong for many riders (LT1 ranges $\sim 65\text{--}85\%$ of CP), and it gates whether the glycolytic tank recovers during tempo, so a bad value materially changes recovery estimates.
- **Several small modeling choices are assumptions, flagged not defended.** The LT1 gate shape ($(LT1-P)/LT1$, linear, vanishing just below LT1 — its slope depends on LT1's absolute value, and it arguably switches off metabolite clearance too abruptly through the tempo band); lumping deficit clearance and glycolytic-tank refill onto the *same* τ_g ; and the spill order (unmet demand spills to glycolytic before PCr). Each is a defensible default, none is derived — named here so a reader weights them accordingly.
- **The deficit D corrupts the display precisely at maximal effort.** When D is large the two bars read slightly full; D grows when power exceeds the (possibly stale) P_{p_max} rate cap, and P_{1s} decays intra-ride with fatigue, so this recurs late in hard efforts — a display *failure mode*, not just an accounting footnote. The field should surface a "recalibrate sprint power" hint when D grows from full tanks (the `rate_limited` flag; §4.3).
- **The product thesis — two bars $\rightarrow$ better pacing — is untested as a human-factors claim.** §6.8 shows the bar is *informative* often enough to matter on interval sessions, but not that a rider *decides better* with it. The minimum next step is a usability check (even $n=1$: race on it, log whether the second bar changed a call) before the display's value is asserted, not just its informativeness.
- **$C_p = 0$ means the usable alactic store is spent, not that muscle PCr is zero.** Real PCr bottoms out around 20–40% of resting at exhaustion; C_p is the usable reserve above that floor, so $R_p = 0$

on the display is "usable punch gone," not a muscle state that never occurs.

- **The compartments are unfalsifiable in-modality — permanently, not "not yet."** Two facts compound: the depletion split is unidentifiable from power (§4.2), and 31P-MRS cannot be run during real cycling (§6.5). Together these are not "untested" — they are "**no in-modality measurement, even in principle, distinguishes the two-compartment model from a one-compartment model with the same aggregate recovery.**" So for cycling the "two tanks" framing is *terminally* a heuristic: it can be motivated, never confirmed in-modality. The only route to compartment-level evidence is **cross-modality transfer** — measure τ_p by knee-extension 31P-MRS (where the magnet works) and assume it transfers — which is itself a nameable, stress-testable assumption, unlike the current no-path-at-all. On *correctness* vs single-tank W'bal there is still no evidence (that needs §6.6). On *usefulness*, v0.5 adds the first real-ride evidence (§6.8): the bar is informative ~45% of interval-ride time and the caps move it materially at decision points — necessary, not sufficient, for the product thesis.
- **Does the depletion-side machinery earn its keep? (Now answered, with data.)** The null model to beat is **a single reserve that depletes exactly as W'bal and recovers bi-exponentially** (fast τ_p
- LT1-gated slow τ_g , split f_p) — the same two bars in recovery, none of the caps/spill/deficit/guards. The answer, from §6.8: the caps **do** move the display on real interval sessions (up to ~25–60 pts at recovery-valley starts), so they are not inert — but they earn their keep in **exactly one regime, the maximal one**, where the tapered ceiling binds and produces the ~23% sprint residual and the post-effort PCr level the recovery law then acts on. Everywhere the ceiling does *not* bind the machinery is inert, and in particular **per-system live consumption (§4.3) is not a reading**: when the ceiling is slack, $\text{take}_p = \Delta \cdot C_p / (C_p + C_g \cdot g)$ contains neither R_p nor R_g — it settles to a fixed $\Delta \cdot f_p / \Delta \cdot (1 - f_p)$ split within ~20 s, i.e. it is $(\text{power} - CP) \times a \text{ clock}$, the power meter rescaled. So: **keep the ceiling (scoped to sprint fidelity), and relabel per-system live consumption as "modelled share," not a measurement** (or drop it). That is the answer to Reviewer 2's S1, drawn from the model's own equations — not "adopt the null wholesale," but "keep the one part that does work and stop claiming the parts that don't." **And there is a second regime the null cannot touch**: cumulative per-system *training load* (§6.9) — the activation ramp makes the split effort-structure-dependent (alactic 76% vs glycolytic 41% PCr), while the null reports a fixed f_p for every session. So the depletion machinery earns its keep in **two** places (maximal sprint fidelity, and training-load partitioning), not zero as the affine-with-W'bal critique implies for pacing alone.
- **Fixed time constants** ignore the documented pH-dependence and bout-to-bout slowing of PCr recovery unless the optional fatigue term is enabled.
- **Hard CP boundary** omits the aerobic slow component and onset kinetics unless the optional aerobic-ramp term is enabled; expect over-attribution to anaerobic tanks in long low-intensity recovery (the same failure mode reported for the EJAP 2023 model).
- **Power is whole-body and mechanical**, not muscle-specific; the model cannot see local fatigue, cadence, or fiber-type effects.
- **Not medical or diagnostic.** This is a training/pacing aid; tank levels are estimates, not measurements of muscle chemistry.
- **Parameters drift** with fitness, fatigue, heat, and altitude; periodic re-testing is required.

8. Conclusion

The single-tank W' model earned its place by being power-native and light enough to run anywhere, but it answers "how much anaerobic work is left?" without answering "in which system?" The dual-tank model proposed here offers a physiologically-motivated split at negligible computational cost: divide W' into a fast, rate-limited, fast-recovering PCr tank and a slow, large, slowly-recovering glycolytic tank; share supra-CP demand between them by **capacity**, cap each by its **peak flux** (PCr tapering with fullness), and refill them with system-specific, intensity-gated laws. It generalizes W' bal in depletion and diverges from it — deliberately — in recovery, it runs at 1 Hz on a Garmin head unit, and it surfaces a two-system decomposition — *punch left* and *dig left* — that is most informative in the transients where pacing decisions concentrate.

The honest bottom line, stated rather than gestured at: **there is still no evidence the two-bar decomposition is more correct than single-tank W' bal** (that needs §6.6, and at the compartment level is unfalsifiable in-modality even in principle — §6.5, §7). What v0.5–0.6 *do* add is **usefulness** arguments: on real interval rides the PCr bar is informative ~45% of the time and the caps move the display up to ~25–60 pts at decision points (§6.8); and — the strongest standalone point — for **training** the cumulative per-system load distinguishes an alactic session (~76% PCr) from a glycolytic one (~41%), information both W' bal and the recovery-only null structurally lack (§6.9). So the depletion machinery is not inert. That resolves the fork more precisely than "ship vs re-architect": **ship the heuristic, keep the ceiling scoped to sprint fidelity, and stop claiming the parts of the machinery the equations show are inert** (per-system live consumption is (power – CP) rescaled, not a reading — §7). The τ_g confound is closed (Ferguson at 20 W $\Rightarrow$ 470; §4.1a), and a genuine tension is now hosted rather than buried: Ferguson's *own* recovery components are ~4× slower than the model's while the aggregate fits, and Ferguson concludes the decomposition cannot be done — the strongest single datum in the paper, and it points at the model's premise. What remains decisive is the **recovery-law head-to-head** (§6.6). The physiological framing has run out of room; only that experiment can move this further.

References

See `docs/literature-review-anaerobic-models.md` for the full annotated bibliography with DOIs and PMIDs. Primary sources for this paper:

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Sourcing note: equations for the incumbent models were cross-checked against PubMed Central full text (PMC7552657, PMC10638188); typeset equations there were image-embedded, so published closed forms were reconstructed from the surviving prose and standard literature. Verify exact coefficients against publisher PDFs before hard-coding.